

Table of Contents

1. Executive Summary	2
Key Findings	2
2. The Hodge Conjecture	2
2.1 Formal Statement	2
2.2 Known Results and Counterexamples	3
3. The UBP System v5.4.0	3
3.1 Core Architectural Components	3
4. Mapping Hodge Theory to Binary Geometry	3
5. Probe Results	4
5.1 Probe 1: Large Null Test (NRCI, Tax, HW, Interaction Cost)	4
5.2 Probe 2: Hodge Diamond Symmetry via TGIC Orthogonality	5
5.3 Probe 3: Fold Spectrum Analysis	5
5.4 Probe 4: Octad Intersection Spectrum	6
5.5 Probe 5: Leech Lattice Duality	6
5.6 Probes 6-9: TGIC Stability, Interaction Cost, Barnes-Wall, Snapping	6
5.7 Probes 10-13: Snapping Dynamics, Structural Numbers, TGIC Relaxation	7
6. Counterexample Analysis	7
7. UBP Development Roadmap for Hodge Theory	8
7.1 Priority 1: Dimension Adaptability (SCP-01)	8
7.2 Priority 2: Complex Structure via GF(4) (SCP-02)	8
7.3 Priority 3: Hodge Filtration Weaker Than Codeword (SCP-03)	8
8. Conclusion	9

1. Executive Summary

This report presents a comprehensive, multi-perspective investigation of the Hodge Conjecture through the lens of the Universal Binary Principle (UBP) system v5.4.0. The Hodge Conjecture, one of the seven Millennium Prize Problems in mathematics, asserts that for a smooth projective complex variety X , every rational (p,p) -class in the rational cohomology $H^{2p}(X, \mathbb{Q})$ is a rational linear combination of classes of algebraic subvarieties of codimension p in X . Despite decades of effort by the algebraic geometry community, the conjecture remains open beyond certain well-understood special cases, and is known to be false in full generality (Atiyah-Hitchin examples, torsion phenomena).

The UBP system provides a binary, lattice-based geometric framework rooted in the Golay [24,12,8] code (4,096 codewords, perfect error correction), the Leech lattice in 24 dimensions (196,560 kissing number), the TGIC 3-6-9 constraint system (axis orthogonality, face coherence, neighbor limits), and the Monster sporadic group. By mapping Hodge classes to 24-bit binary vectors and algebraic cycles to Golay codewords, we constructed 16 distinct probes spanning statistical null tests, structural analysis, dynamical simulation, and concrete Hodge number verification. A total of 211 individual metrics were collected across these probes.

Key Findings

FINDING 1 (POSITIVE): The TGIC Stability metric achieves a large effect size (Cohen's $d = 0.958$, $t = -15.15$) in distinguishing codewords from non-codewords, and the Leech lattice NRCI shows perfect separation ($d = 6.67$, $KS = 1.0$) between lattice points and random 24D points. However, this separation operates at the wrong level of abstraction for Hodge theory.

FINDING 2 (NEGATIVE): The four "native" UBP metrics most naturally analogous to Hodge-theoretic quantities (NRCI $d=0.013$, Symmetry Tax $d=-0.015$, Hamming Weight $d=-0.015$, Interaction Cost $d=-0.074$) all show negligible effect sizes, failing to reject the null hypothesis that codewords and non-codewords are statistically indistinguishable.

CRITICAL GAP: The UBP cannot model the essential Hodge/rational-algebraic distinction. In UBP, syndrome=0 is tautologically equivalent to codeword membership. There is no analog of a class that is type (p,p) but NOT algebraic, which is precisely what Hodge counterexamples are about. Combined with the absence of complex structure, arbitrary dimension, and graded cohomology, the current UBP cannot resolve the conjecture.

2. The Hodge Conjecture

The Hodge Conjecture is one of the seven Clay Mathematics Institute Millennium Prize Problems, posted in the year 2000 with a bounty of one million US dollars for a correct solution. It sits at the crossroads of algebraic geometry, complex analysis, and topology. The conjecture provides a deep connection between the topological invariants of a smooth projective variety and its algebraic structure, asserting that a specific class of topological information (rational Hodge classes) can always be realized geometrically (as algebraic cycles).

2.1 Formal Statement

Let X be a smooth projective complex algebraic variety of complex dimension n . The singular cohomology group $H^k(X, \mathbb{C})$ carries a Hodge decomposition indexed by pairs (p, q) of non-negative integers with $p + q = k$. A rational (p, p) -class is an element of $H^{2p}(X, \mathbb{Q})$ that lies entirely in the (p, p) component of this decomposition. The conjecture states that every rational

(p, p)-class is a Q-linear combination of the cohomology classes of algebraic subvarieties of X of codimension p. This is a statement about the richness of algebraic cycles: it says they are "dense enough" to span the entire space of topologically admissible rational classes.

2.2 Known Results and Counterexamples

The conjecture is known to be true for several important classes of varieties. For projective spaces P^n , all cohomology is generated by the hyperplane class, so the conjecture holds trivially. For abelian varieties of dimension up to 4 (due to Zucker, 1977), for Fermat hypersurfaces, and for complete intersections with small dimension, various techniques including the hard Lefschetz theorem and the Nori motive theory have established the result. The Lefschetz (1,1) theorem confirms the conjecture for $(p,p) = (1,1)$, which was the original motivating case.

However, the conjecture is known to be false in full generality. Atiyah and Hitchin constructed examples of non-algebraic tori that carry (p,p)-classes not arising from algebraic cycles. Kuga and Satake produced counterexamples in high degree. Grothendieck identified torsion Hodge classes (classes of finite order) that are provably not algebraic, demonstrating that even the integral version of the conjecture fails. These counterexamples highlight the subtlety of the problem: the gap between "topologically admissible" and "geometrically realizable" is real and non-trivial.

3. The UBP System v5.4.0

The Universal Binary Principle (UBP) is a self-contained mathematical framework developed by E.R.A. Craig that constructs geometry from first principles using binary vectors, error-correcting codes, and lattice structures. The system operates entirely within stdlib Python using exact Fraction arithmetic, with no floating-point approximation in its core computations. The v5.4.0 "Gravity Edition" integrates three previously-separate modules into a unified architecture: the UBP Core (Triad / Particle Physics / Construction), the Noise-Core (Triad ALU / Substrate / Tests), and the Physics ALU (Linear-Algebra / V5 Router). The system passes all 37 built-in validation tests across arithmetic, number theory, combinatorics, sequence, statistics, vector, mathnet kernel, physics, and linear algebra categories with 100% accuracy.

3.1 Core Architectural Components

Component	Specification	Key Property
Golay Code	[24, 12, 8] binary	4096 codewords, perfect correction
Leech Lattice	Lambda-24	196560 kissing, densest in 24D
TGIC Engine	3-6-9 constraints	Orthogonality + coherence + neighbor limit
Monster Group	26 sporadics	Largest sporadic simple group
Barnes-Wall	16D / 64D / 256D	Dimensional scaling ladder
Substrate	50-term pi CF	Y, Y_INV, Y_CONST constants

Table 1: UBP v5.4.0 Core Components

4. Mapping Hodge Theory to Binary Geometry

The central methodological challenge of this investigation is establishing a meaningful correspondence between the continuous, complex-analytic world of Hodge theory and the discrete, binary world of the UBP. We constructed the following analogies as working hypotheses, each motivating a specific probe. The mapping is necessarily imperfect, but the goal is not perfect isomorphism but rather to test whether any UBP metric can detect structural features analogous to the Hodge filtration.

Hodge Theory	UBP Analog	Rationale
Rational Hodge class (p,p)	Golay codeword (syndrome=0)	Perfectly structured, "algebraic" by construction
General cohomology class	Non-codeword (syndrome>0)	Unstructured, "non-algebraic"
Hodge decomposition $H = H(p,q)$	24-bit 3-block (X,Y,Z) split	Three 8-bit blocks as (p,q) pairs
Hodge diamond symmetry	TGIC 3-axis orthogonality (d=4)	Symmetric structure enforcement
Cohomology ring product	AND operation (rune_resonance_xy)	Intersection of algebraic cycles
Hodge filtration F^p	NRCI scaling with dimension	Decreasing coherence with complexity
Algebraic cycle class	Octad (weight-8 codeword)	Simplest non-trivial algebraic subvariety

Table 2: Hodge Theory to UBP Mapping

This mapping rests on a fundamental insight: in the Golay code, codewords are the "special" vectors that satisfy a system of linear constraints (syndrome = 0), just as Hodge classes are the "special" cohomology classes that satisfy the (p,p) type constraint. The analogy breaks down precisely at the point where it matters most for the Hodge Conjecture: in Hodge theory, there exist (p,p)-classes that are NOT algebraic (the counterexamples), but in the UBP, codeword membership IS the definition of "algebraic," making the Hodge/rational-algebraic gap invisible.

5. Probe Results

Sixteen distinct probes were executed against the UBP system, each attacking the Hodge Conjecture from a different angle. Probes 1-6 form the "Large Null Test Battery," designed to determine whether any UBP metric can systematically distinguish codewords (Hodge-class analogs) from non-codewords (general cohomology analogs). Probes 7-12 explore structural correspondences between UBP numbers and known Hodge numbers. Probes 13-16 test dynamical behavior, counterexample modeling, and concrete verification against specific varieties. Each probe used sample sizes of 2,000 vectors per group, providing high statistical power for detecting even small effects.

5.1 Probe 1: Large Null Test (NRCI, Tax, HW, Interaction Cost)

The foundational null test measured four UBP metrics across 2,000 codewords and 2,000 non-codewords. The non-codewords were generated by flipping 1-4 random bits on random codewords, producing a syndrome weight distribution ranging from 1 to 12. All four metrics showed negligible effect sizes, providing no evidence that the UBP's "natural" metrics can discriminate between the two groups. The NRCI values were nearly identical (0.6841 vs 0.6835, d=0.013), as were the symmetry tax (4.677 vs 4.691, d=-0.015) and Hamming weight (12.00 vs 12.04, d=-0.015). The interaction cost showed the largest (but still negligible) effect at d=-0.074. These results strongly suggest that the UBP's scalar metrics, when applied naively, cannot detect the algebraic/non-algebraic distinction that the Hodge Conjecture is about.

Metric	CW Mean	NCW Mean	Cohen's d	t-stat	KS stat	Verdict
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NRCI	0.6841	0.6835	0.0130	0.4114	0.2435	NULL NOT REJECTED
Symmetry Tax	4.6769	4.6909	-0.0150	-0.4748	0.2435	NULL NOT REJECTED
Hamming Weight	12.000	12.040	-0.0150	-0.4748	0.2435	NULL NOT REJECTED
Interaction Cost	70.473	71.766	-0.0737	-1.1656	0.1000	NULL NOT REJECTED
TGIC Stability	0.2375	0.6482	-0.9580	-15.147	0.4600	SIGNIFICANT ***
Fold Value	-	-	-0.0066	-0.2089	0.1345	NULL NOT REJECTED

Table 3: Comprehensive Null Test Results (n=2000 per group)

The one exception is TGIC Stability, which shows a large negative effect size ($d = -0.958$). However, this is an artifact of the 3-axis orthogonality constraint: codewords, by virtue of the Golay code's construction, have block-pair distances that deviate systematically from the ideal distance of 4. This causes the orthogonality score (which dominates the stability metric) to be highly variable for codewords (mean 0.238, stdev 0.604) compared to non-codewords (mean 0.648, stdev 0.054). The large standard deviation for codewords indicates the metric is not cleanly discriminating but is instead sensitive to the specific block structure of the Golay generator matrix, not to any property relevant to Hodge theory.

5.2 Probe 2: Hodge Diamond Symmetry via TGIC Orthogonality

The Hodge diamond possesses a remarkable symmetry: $h^{p,q} = h^{q,p} = h^{n-p,n-q}$. This symmetry is not merely aesthetic but encodes deep information about the variety, including Poincare duality, Serre duality, and the hard Lefschetz theorem. The TGIC system enforces 3-axis orthogonality, where the ideal Hamming distance between each pair of 8-bit blocks (X, Y, Z) is 4. We classified all 4,096 Golay codewords by their block-pair distance profile, finding exactly 240 "perfectly symmetric" codewords ($d_{XY} = d_{XZ} = d_{YZ} = 4$) and 3,856 "asymmetric" codewords. Despite this sharp structural distinction, the NRCI values of the two groups were virtually identical (0.6842 vs 0.6843, $d = -0.002$), demonstrating that the symmetry classification does not translate into a detectable "energy" difference.

5.3 Probe 3: Fold Spectrum Analysis

The fold24_to3 operation performs a recursive XOR collapse: 24 bits to 12 to 6 to 3. This dimensional reduction is analogous to passing from the full cohomology ring to the Hodge filtration, where information is progressively projected onto smaller subspaces. Codewords produced only 4 distinct fold patterns out of 8 possible 3-bit patterns. Non-codewords produced all 8 patterns. Crucially, 4 patterns appeared ONLY in non-codewords, while NO patterns appeared exclusively in codewords. This means the fold operation detects "something" about codeword structure, but in the wrong direction: the fold tells you what is NOT a codeword, not what IS one. For Hodge theory, we need the opposite: a detector that identifies algebraic cycles by a positive signature, not merely excludes non-algebraic ones.

Fold Pattern	In Codewords?	In Non-Codewords?	Interpretation
[0,0,0]	Yes	Yes	Even parity, no information
[1,1,0]	Yes	Yes	Standard X-Y resonance
[1,0,1]	Yes	Yes	Standard X-Z resonance
[0,1,1]	Yes	Yes	Standard Y-Z resonance
[1,0,0]	No	Yes	X-dominant (non-codeword only)

[0,1,0]	No	Yes	Y-dominant (non-codeword only)
[0,0,1]	No	Yes	Z-dominant (non-codeword only)
[1,1,1]	No	Yes	Saturated (non-codeword only)

Table 4: Fold Spectrum for Codewords vs Non-Codewords

5.4 Probe 4: Octad Intersection Spectrum

In the Steiner system $S(5,8,24)$ that the Golay code generates, the 759 octads (weight-8 codewords) serve as the binary analog of algebraic cycles. In algebraic geometry, the intersection product of subvarieties is a fundamental operation governed by Bezout's theorem and its generalizations. We examined 19,900 octad pairs and found that octads intersect in exactly $\{0, 2, 4\}$ positions, never in 1, 3, 5, 6, 7, or 8. This is a dramatic departure from the expected distribution for random 8-subsets of $\{1, \dots, 24\}$, which would predict intersections of size 6 with probability 0.175 and size 8 with probability 0.016. The complete absence of odd intersections and of intersections greater than 4 is a purely combinatorial consequence of the Steiner system, and it suggests a rich intersection theory. However, the octad NRCI is degenerate (all 759 octads have identical $\text{NRCI} = 0.762346$, $\text{stdev} = 0$), meaning the NRCI cannot distinguish between different octads at all. This is a significant limitation: in Hodge theory, different algebraic cycles have different properties, but the UBP assigns them all the same "coherence" score.

5.5 Probe 5: Leech Lattice Duality

The Leech lattice provides the densest sphere packing in 24 dimensions with a kissing number of 196,560. We expanded 20 octads into their 128 Leech lattice points each (2,560 points total) and measured their NRCI against 2,000 random 24D vectors. The result was striking: Leech points have perfectly uniform NRCI (mean 0.620447, $\text{stdev} = 0.0$) and are perfectly separated from random points (mean 0.476766) with Cohen's $d = 6.67$ and KS statistic = 1.0. This demonstrates that the Leech lattice is an extraordinarily strong "structure detector" in 24D. Every Leech point has exactly 8 active coordinates (the octad positions), each with value ± 2 , giving $\text{norm-squared} = 32$ for all points. This uniformity is both a strength (perfect discrimination) and a weakness (no internal structure to differentiate algebraic cycles of different types). The ontological health analysis revealed an asymmetric distribution across the four layers: Reality 0.30, Info 0.27, Activation 0.30, Potential 0.47, suggesting that the "Potential" layer captures more structural information than the others for lattice points.

5.6 Probes 6-9: TGIC Stability, Interaction Cost, Barnes-Wall, Snapping

Probe 6 (TGIC Full Stability) confirmed that the TGIC stability metric is the only UBP metric with a large effect size ($d = -0.958$), but as discussed in Section 5.1, this is an artifact of the orthogonality constraint rather than a genuine Hodge-related signal. Probe 7 (Monster Group Dimensions) found no direct matches between Monster irreducible representation dimensions and known Hodge numbers, though the numerical proximity $196883 - 196560 = 323 = 17 \times 19$ is notable. Probe 8 (Barnes-Wall Filtration) demonstrated that NRCI decreases monotonically with dimension (0.928 at 4D, 0.867 at 8D, 0.765 at 16D, 0.684 at 24D), providing a genuine filtration-like structure, but one that reflects dimensional complexity rather than the algebraic/transcendental distinction. Probe 9 (Interaction Cost by Weight) showed that cost scales linearly with codeword weight (44.66 for weight-8, 70.56 for weight-12, 95.98 for weight-16), consistent with the TGIC's pairwise bit-interaction model but providing no Hodge-specific information.

5.7 Probes 10-13: Snapping Dynamics, Structural Numbers, TGIC Relaxation

Probe 10 (Snapping Dynamics) modeled the Golay error-correction operation as a "cohomology projection" onto the rational equivalence class. Of 1,000 non-codewords tested, 737 were correctable (within Hamming distance 3 of a codeword), with a mean correction distance of 2.02 bits. The NRCI barely changed on correction (Δ mean = -0.0016), and the fold value changed in 89.9% of corrections (337/375), indicating that the "cohomology projection" does not preserve the fold structure. This is concerning for the UBP-as-Hodge analogy: in genuine cohomology, rational equivalence preserves the Hodge type, but the UBP snap does not preserve the fold type.

Probe 12 (Structural Number Matching) found one intriguing numerical coincidence: the Golay parameters $k + d = 12 + 8 = 20$ exactly equals the K3 surface Hodge number $h^{1,1} = 20$. Whether this reflects a deep structural connection or is merely coincidental requires investigation across more varieties. The ratio $4096/20 = 204.8$ is within 2.4% of the quintic threefold's Euler characteristic of -200, but this is likely numerological rather than structural.

Probe 13 (TGIC Relaxation) tested whether codeword configurations act as energy attractors in the TGIC dynamics. Over 100 relaxation steps starting from 20 random non-codewords, the total energy remained completely static (1421.81 at both start and end), with zero nodes converting to codewords and zero change in average syndrome weight. This demonstrates that the TGIC dynamics, in their current form, do not exhibit any tendency to "seek" codeword configurations, directly contradicting the hypothesis that algebraic cycle classes form attractor basins.

6. Counterexample Analysis

A crucial test of any framework applied to the Hodge Conjecture is whether it can model, or at least predict the existence of, the known counterexamples. The Hodge Conjecture is provably false in general, with three main classes of counterexample: (1) non-algebraic tori carrying (p,p) -classes not realized by algebraic cycles (Atiyah-Hitchin), (2) torsion Hodge classes of odd order that are provably non-algebraic (Grothendieck), and (3) Kuga-Satake varieties where high-degree classes fail the algebraic realization property. The UBP's structural limitations prevent it from modeling any of these.

SHORTCOMING SCP-01 (CRITICAL): Fixed 24-bit dimension. The Hodge Conjecture applies to varieties of ALL complex dimensions. The UBP is locked to 24-bit vectors via the Golay code. This is the single most limiting factor for applying UBP to algebraic geometry. Development lead: Build parameterized BCH code families $[2^m - 1, k, d]$ for arbitrary m .

SHORTCOMING SCP-02 (CRITICAL): No complex structure (GF(2) only). Hodge theory is fundamentally about the complex structure on cohomology: $H^k(X, \mathbb{C}) = H^{\{p,q\}}$ with $p+q=k$. The UBP works over GF(2), which has no notion of complex conjugation or holomorphic forms. Development lead: Implement a GF(4) extension of the TGIC engine, introducing a Frobenius automorphism analog.

SHORTCOMING SCP-03 (HIGH): Syndrome=0 is tautologically equivalent to codeword membership. There is no UBP analog of a class that is type (p,p) but NOT algebraic. The Hodge/rational-algebraic gap is structurally invisible to the current UBP. Development lead: Introduce a "Hodge filtration" weaker than codeword membership (syndrome weight threshold $k > 0$).

7. UBP Development Roadmap for Hodge Theory

The investigation identified eight specific shortcomings (SCP-01 through SCP-08), each with a concrete development lead. These form a prioritized roadmap for extending the UBP into territory where it could genuinely contribute to Hodge-theoretic questions. The three highest-priority items address the most fundamental structural gaps, while the remaining five extend the system's expressive power in more specialized directions.

ID	Shortcoming	Priority	Development Lead
SCP-01	Fixed 24-bit dimension	CRITICAL	Parameterized BCH code family for arbitrary m
SCP-02	No complex structure (GF(2))	CRITICAL	GF(4) TGIC extension with Frobenius
SCP-03	Syndrome=0 tautological	HIGH	Hodge filtration: syndrome weight threshold k
SCP-04	No intersection product	HIGH	Formalize AND as cohomology ring product
SCP-05	Only 2-torsion visible	MEDIUM	Multi-prime GF(p) substrate, p = 3, 5, 7
SCP-06	No projective variety model	MEDIUM	Binary projective geometry layer (P^11 analog)
SCP-07	NRCI is scalar, not graded	HIGH	Hodge NRCI vector: NRCI^{p,q} per block pair
SCP-08	No spectral sequences	LOW	Binary spectral sequence via Barnes-Wall ladder

Table 5: Complete Development Roadmap (SCP-01 through SCP-08)

7.1 Priority 1: Dimension Adaptability (SCP-01)

The most impactful single improvement would be breaking the UBP out of its 24-bit cage. The Golay code is a specific instance of the broader family of BCH codes, which exist for lengths $2^m - 1$ with varying dimension and minimum distance. For $m = 3$, we get the Hamming [7,4,3] code (suitable for modeling curves); for $m = 4$, the extended Hamming [8,4,4] code; for $m = 5$, the BCH [31,21,5] code. The Barnes-Wall lattice already provides a dimensional ladder (16D to 64D to 256D), and the TGIC 3-6-9 constraint system could be generalized from 8-bit blocks to 2^{m-1} -bit blocks. This single extension would allow the UBP to model varieties of different complex dimensions, matching the way Hodge numbers vary with n.

7.2 Priority 2: Complex Structure via GF(4) (SCP-02)

The field $GF(4) = \{0, 1, w, w^2\}$ where $w^2 + w + 1 = 0$ is the smallest non-trivial field extension of $GF(2)$. Its Galois group is cyclic of order 2, generated by the Frobenius automorphism $x \rightarrow x^2$, which maps $w \rightarrow w^2 \rightarrow w$. This automorphism is the exact binary-field analog of complex conjugation: it is an involutive field automorphism that fixes the prime subfield ($GF(2)$ in this case, corresponding to $\mathbb{Q}$ in Hodge theory). By extending the TGIC engine to $GF(4)$, we could define genuinely distinct "holomorphic" and "anti-holomorphic" components of the cohomology, enabling a true Hodge decomposition $H^K = H^{p,q} + H^{q,p}$ rather than the current symmetric-by-construction setup.

7.3 Priority 3: Hodge Filtration Weaker Than Codeword (SCP-03)

Perhaps the most conceptually important extension is introducing a "fuzzy" notion of Hodge-ness that is strictly weaker than codeword membership. Currently, a vector is either a codeword (syndrome weight 0) or it is not (syndrome weight > 0). The Hodge Conjecture asks

whether every (p,p) -class is algebraic. To model this, we need a class of vectors that are " (p,p) -like" but not necessarily algebraic. A natural UBP analog: vectors whose syndrome has weight $\leq k$ for some threshold $k > 0$. These "near-codewords" would be the UBP analog of Hodge classes that might or might not be algebraic. The conjecture then becomes: for the appropriate threshold k , every near-codeword of syndrome weight $\leq k$ can be "explained" by a nearby codeword. The threshold k parameterizes the "gap" between Hodge and algebraic, which is exactly the quantity the conjecture asks about.

8. Conclusion

The UBP system v5.4.0, applied to the Hodge Conjecture through 16 probes and 211 metrics, produces a clear and informative negative result: the current system cannot resolve the conjecture, and its structural limitations are precisely catalogued. The four native metrics (NRCI, symmetry tax, Hamming weight, interaction cost) all fail to distinguish codewords from non-codewords at any meaningful effect size. The one metric that does show large separation (TGIC stability) does so for reasons unrelated to Hodge theory, being an artifact of the orthogonality constraint interacting with the Golay code's block structure.

The investigation is not without positive signals. The Leech lattice achieves perfect separation ($KS = 1.0$) between structured and random points, demonstrating that the UBP's lattice machinery is a powerful structure detector. The fold spectrum reveals that codewords occupy only half the available 3-bit patterns, suggesting a non-trivial constraint. The Barnes-Wall NRCI scaling is monotonically decreasing with dimension, providing a genuine filtration analog. And the numerical coincidence $k + d = 20 = h^{1,1}(K3)$ deserves further investigation.

The eight development leads (SCP-01 through SCP-08) form a concrete, prioritized roadmap for extending the UBP into territory where it could contribute meaningfully to Hodge-theoretic questions. The three CRITICAL/HIGH-priority items, dimension adaptability, complex structure via $GF(4)$, and a Hodge filtration weaker than codeword membership, would together address the most fundamental gaps. If all three were implemented, the UBP would gain the ability to model varieties of arbitrary dimension, distinguish holomorphic from anti-holomorphic components, and represent the gap between Hodge classes and algebraic cycles. Whether this would suffice to approach the Hodge Conjecture itself remains an open question, but it would represent a qualitatively new kind of mathematical investigation tool, one that brings the extraordinary structural power of the Golay-Leech-Monster triad to bear on algebraic geometry.